

Financial Mathematics

(MATH35400/MATHM5400)

Problem Class 4 – 13 February 2026 ¹

1. (2019 exam, part of Q2) Consider a model with $N = 1$ stock over $T = 2$ periods and $K = 4$ possible states. The price process is:

ω_k	$t = 0$	$t = 1$	$t = 2$
ω_1	$S_0 = 5$	$S_1 = 8$	$S_2 = 10$
ω_2	$S_0 = 5$	$S_1 = 8$	$S_2 = 6$
ω_3	$S_0 = 5$	$S_1 = 4$	$S_2 = 6$
ω_4	$S_0 = 5$	$S_1 = 4$	$S_2 = 2$

Assume that a bank account is available with interest rate r in both periods.

- (a) State the difference between an American call option and a European call option on the same stock, with the same exercise data and exercise price.
 - (b) Write down the definition of a σ -algebra associated with a finite sample space Ω .
 - (c) Write down the filtration generated by the price process in the table above.
 - (d) By considering conditional probabilities, find the range of values of r for which risk neutral measures exist.
2. (2020 exam, level M, Q2b) Consider a multi-period model with $N = 1$ risky security and a finite probability space $\Omega = \{\omega_1, \dots, \omega_K\}$. For each state $\omega \in \Omega$, time point $t = 0, 1, \dots, T$, let $S(t)(\omega)$ denote the price of the security at time point t in state ω . We assume that there is a bank account with interest rate $r = 0$.

Let us assume that the price process satisfies

$$S(0) = 1, \quad S(t) = \begin{cases} S(t-1) \exp(X(t-1)), & \text{if } S(t-1) > \epsilon \\ S(t-1) \exp(X(t-1) + 2), & \text{if } S(t-1) \leq \epsilon \end{cases}, \quad t = 1, \dots, T$$

where $\epsilon \in (0, 1)$ is a constant and $X = (X(t), t \geq 0)$ is a sequence of i.i.d. random variables such that each $X(t)$ takes each of the values 1 and -1 with probability $1/2$.

Construct an arbitrage opportunity for this model and verify all the conditions which define such an opportunity. Hint: for a complete answer, you will need to state appropriate assumptions on ϵ and T .

¹©University of Bristol 2026. This material is copyright of the University of Bristol unless explicitly stated otherwise. It is provided exclusively for educational purposes at the University and is to be downloaded or copied for your private study only.